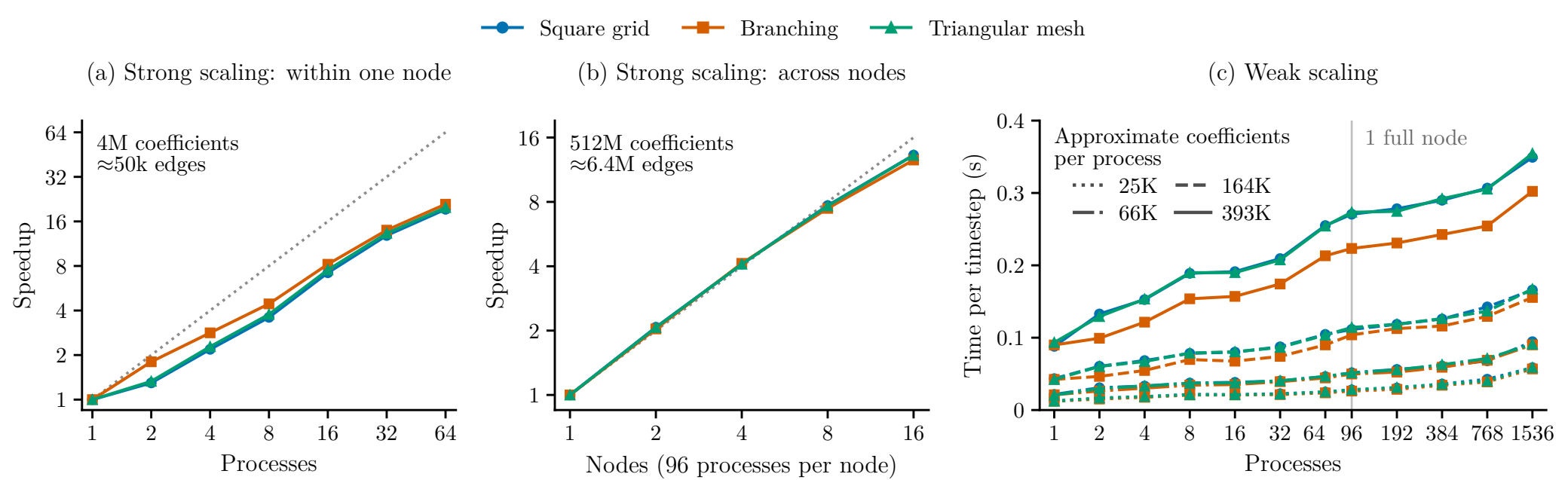

Square grid
Branching
Triangular mesh
(a) Strong scaling: within one node
(b) Strong scaling: across nodes
(c) Weak scaling
4M coefficients
≈50k edges
512M coefficients
≈6.4M edges
Approximate coefficients per process
25K
66K
164K
393K
1 full node
Speedup
Time per timestep (s)
Processes
Nodes (96 processes per node)